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FPMBGCN : Feature pyramid-based multi-branch graph convolutional networks for epileptic seizure detection

Zhongfei Zhang ^a, Jialong Lai ^a  , Xu Zhang ^a, Feifei Cui ^a, Zheng Hu ^b, Yanping Feng ^b, Ying Wan ^b

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Highlights

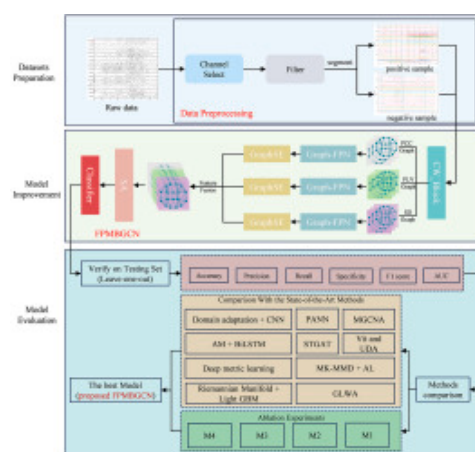
- FPMBGCN combines multi-branch GCN and feature pyramid for cross-patient EEG seizures.
- Constructs three graph structures to represent the topology of EEG signals.
- Achieves state-of-the-art classification performance on the CHB-MIT dataset.

Abstract

Epilepsy is a prevalent neurological disorder, and the automatic detection of seizure periods plays a vital role in clinical diagnosis and treatment. However, significant variability in seizure patterns across patients poses substantial challenges for cross-patient detection, including pattern heterogeneity and noise interference. Traditional convolutional neural networks struggle to model the topological dependencies between EEG channels, while existing graph neural network (GNN) approaches, which typically rely on a single graph input, exhibit limited capacity in capturing inter-channel topological relationships. To address these limitations, this paper proposes the feature pyramid-based multi-branch graph convolutional network (FPMBGCN) for epileptic seizure

detection using multi-channel EEG signals. The method constructs three complementary graph-structures: pearson correlation coefficient (PCC), phase-locking value (PLV), and spatial distance (SD) graphs—to extract shared topological features across patients. It further incorporates a channel weight block to adaptively emphasize critical channels, a graph feature pyramid to capture multi-scale information while suppressing noise, and the graph squeeze-and-excitation (GraphSE) module alongside a self-attention mechanism to enhance the representation of spatial and temporal dynamics. The performance of the proposed method is evaluated on both the CHB-MIT dataset and the Siena scalp EEG dataset with leave-one-out cross-validation. For the CHB-MIT dataset, the proposed FPMBGCN achieves an average accuracy of 92.21%, a precision of 94.50%, a recall of 90.03%, an F1 score of 78.59%, and an F1 score of 91.64% in cross-patient detection. On the Siena dataset, it achieves an accuracy of 77.94%, a precision of 76.25%, a recall of 81.37%, a specificity of 74.41%, an F1 score of 78.59%.

Graphical abstract



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Introduction

Epilepsy is a common neurological disorder, characterized by sudden and paroxysmal abnormal electrical discharges in brain neurons, and is classified as a chronic non-communicable disease [1]. Epileptic seizures originate from an imbalance between excitatory and inhibitory synaptic activity in the brain, resulting in abnormal electroencephalographic (EEG) patterns, which typically manifest as limb convulsions, loss of consciousness, sensory disturbances, or behavioral changes [2]. The diversity and unpredictability of these symptoms make epilepsy a disease that has a profound impact on patients' quality of life. According to statistical data from the World Health Organization (WHO), approximately 50 million people worldwide are affected by epilepsy [3], and this number continues to grow at a rate of millions of new cases each year. Frequent epileptic seizures pose serious threats to both the physical and mental health of patients, potentially leading to memory impairment, cognitive decline, and overall functional disability, and in severe cases, may even be life-threatening. Furthermore, epilepsy places a substantial medical and economic burden on patients' families and society.

EEG signals serve as the primary basis for the study and diagnosis of epilepsy and are commonly divided into four stages according to the seizure progression: interictal, preictal, ictal, and postictal [4]. Among these stages, accurate identification of the ictal period is particularly critical, as it is directly associated with the clinical manifestations of epilepsy and the timing of therapeutic interventions. Traditionally, epilepsy diagnosis has relied on multiple medical imaging modalities, including computed tomography (CT), magnetic resonance imaging (MRI), and high-density EEG [5]. Among these techniques, EEG is highly valued for its excellent temporal resolution. Although its spatial resolution is relatively limited, EEG offers irreplaceable advantages in capturing the dynamic characteristics of abnormal neuronal discharges [6], [7] and has been widely recognized as one of the most effective clinical tools for epilepsy diagnosis. Epileptic seizure detection aims to distinguish whether EEG signals correspond to the ictal period. In conventional clinical practice, this task is primarily performed by neurologists and experienced clinicians through visual inspection and manual annotation of EEG recordings [8]. However, this approach faces several challenges. First, the onset time and duration of epileptic seizures are highly unpredictable, making the identification of ictal segments from large-scale EEG data both labor-intensive and time-consuming. Second, manual interpretation is inherently subjective and strongly dependent on clinicians' experience and attention levels, which can result in low detection accuracy and high misdiagnosis rates [9]. Such limitations not only restrict the accuracy of epilepsy diagnosis but also, to some extent, delay the treatment process for patients. Therefore, the use of automated methods for EEG signal analysis is crucial for simplifying and accelerating seizure detection and has become the main focus of most existing studies.

Due to the typical duration of individual EEG recordings, which can last hours to tens of hours, researchers often segment EEG signals into short time windows and classify the signals within each window to determine whether the data corresponds to the ictal period. Mainstream research employs machine learning or deep learning methods to address this problem [10], [11]. However, most previous studies focus on single-patient training and testing strategy, in which models are trained and tested using data from the same patient, with a portion held out as a test set. This setting inevitably limits the practical applicability of such approaches. Specifically, classification models show high detection performance only for patients whose data they have seen, with generalization declining for unseen patients. In contrast, the cross-patient setting strictly separates patients used for training and testing, where models are trained on EEG data from multiple patients and evaluated on patients who are not involved in the training process. Cross-patient experiments are designed to assess the performance of models trained on multi-patient data when applied to EEG recordings from previously unseen patients [12]. In practice, cross-patient research more closely resembles real-world scenarios, which is vital for clinical drug treatment, as clinical applications typically favor general models that work across different patients and require robust generalization to adapt to new patient data. However, traditional convolutional neural networks (CNN) are unable to effectively capture the topological relationships among EEG channels, limiting their ability to represent complex spatial structures. Additionally, although graph neural network (GNN) typically model EEG data using a single graph structure, which can encode functional connectivity, such a representation struggles to simultaneously incorporate spatial positional information and dynamic dependencies, especially patient-invariant features shared across subjects. As a result, model performance in cross-patient scenarios may deteriorate significantly.

Compared with single-patient studies, cross-patient research is more difficult but has greater potential for improvement. This is because there is a lot of variability in seizure patterns among different patients. In addition, in EEG recordings, ictal data are far less abundant than interictal data, resulting in a severe class imbalance between positive and negative samples, which degrades the performance of classification models. This issue becomes even more pronounced in cross-patient experimental settings. For example, in the latest epilepsy detection studies [13], models tested on the CHB-MIT dataset achieved an average accuracy of 99.32% and an average recall of 98.74% in single-patient tests. However, in cross-patient tests, the performance dropped to an average accuracy of 89.45% and a recall of 84.72%.

The main contributions of this paper are as follows:

- **Multi-branch graph convolutional network:** This study proposed a multi-branch graph convolutional network framework that constructs functional connectivity among EEG channels using the pearson correlation coefficient and the phase-locking value (PLV), while simultaneously incorporating Euclidean distance to capture spatial positional information between channels. This design enables effective extraction of patient-invariant features and significantly improves generalization performance in cross-patient seizure detection.
- **Channel weight block design:** A channel weight block is introduced to automatically assign adaptive weights to multi-channel EEG signals. By suppressing channels that are less relevant to epileptic seizures, the proposed mechanism reduces the influence of non-informative channels on classification performance.
- **Graph feature pyramid module (GraphFPN):** The introduction of a graph feature pyramid (GraphFPN) adaptively extracts multi-scale graph features through pooling and unpooling operations. Feature fusion is achieved using weighted residual (ReZero) operations, which reduce noise while preserving critical features, and improves the ability of the model to represent complex graph structures.
- **Graph squeeze-and-excitation (GraphSE) Module:** The proposed GraphSE module incorporates the squeeze-and-excitation mechanism to adaptively modify the weights of graph structures at various scales, enhancing feature mappings during global pooling and emphasizing critical granularity, thereby boosting classification performance.
- **Self-attention mechanism:** A self-attention mechanism is incorporated to compute attention scores across different input graph structures, enabling the model to capture dependencies among graph representations and further enhance the discriminative power of local features.

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Section snippets

Related work

EEG-based epileptic seizure detection has long been a popular research topic. Early studies concentrated on feature engineering, extracting time-domain, frequency-domain, or time–frequency features from EEG signals for classification. These feature extraction methods can obtain relevant information from raw data to better understand what the model is focused on at any moment, thereby improving classification accuracy. For example, Tian et al. [14] used fast fourier transform (FFT) and wavelet ...

Materials

In this section, we explain the methodology for cross-patient epilepsy detection and detail the data-processing pipeline. This study uses the CHB-MIT dataset and Siena Scalp dataset for experiments, with EEG signals preprocessed through channel selection, filtering, and data segmentation to ensure data quality and practical model training. ...

Methods

To address the challenges outlined above, we introduce a feature pyramid-based multi-branch graph convolutional network (FPMBGCN) to extract features and detect epileptic seizures from multi-channel EEG signals. By integrating multiple graph structures, a channel weight block, a graph feature pyramid, a GraphSE module, and a self-attention mechanism, the model improves the representation of complex spatio-temporal EEG features, leading to better performance in cross-patient epilepsy detection ...

Training setting

To thoroughly evaluate the proposed FPMBGCN, we trained both cross-patient and single-patient models. In the cross-patient model, data from multiple patients were used for training, with one patient's data kept aside as the test set. Because interictal signals are more common than ictal signals, the number of interictal and ictal samples was balanced at a 1:1 ratio during training. The model was retrained 24 times, with each patient serving as the test set once, to calculate the average ...

Comparison of performance

In this study, we propose a novel epilepsy detection method based on a graph convolutional network for multi-channel EEG signals, utilizing three graph representations: pearson correlation coefficient, PLV, and spatial distance. Comparative experiments further demonstrate the performance advantages of the FPMBGCN model in epileptic seizure detection tasks. By comparing

our results with previous studies, we gain a comprehensive understanding of the model's performance in cross-patient seizure ...

Conclusion

In this study, our goal was to improve the model's classification performance on new patient data, focusing on extracting common features across patients. We proposed a feature pyramid-based multi-branch graph convolutional network (FPMBGCN), which converts multi-channel EEG signals into three separate graph structures, integrating channel weight, a graph feature pyramid, a GraphSE module, and a self-attention mechanism to enhance spatio-temporal feature extraction and classification ...

CRedit authorship contribution statement

Zhongfei Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Visualization, Writing – original draft . **Jialong Lai:** Supervision, Resources, Funding acquisition, Writing – review & editing . **Xu Zhang:** Data curation, Validation, Visualization . **Feifei Cui:** Writing – review & editing. **Zheng Hu:** Methodology, Investigation, Funding acquisition, Writing – review & editing . **Yanping Feng:** Data curation, Writing – original draft . **Ying Wan:** Investigation, Validation, Formal ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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